Do Sparse Autoencoders Recover Biophysical Features in Pretrained Cardiac Foundation Models? An *in silico* Ground-Truth Case Study

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Abstract

Pretrained cardiac foundation models demonstrate strong clinical diagnostic performance on multi-lead electrocardiograms (ECGs), yet the mechanistic grounding of their latent representations remains largely unverifiable in human cohorts where intracellular cardiac states cannot be invasively recorded. We address this verification bottleneck by leveraging an *in silico* ground-truth cohort of 14,065 forward-simulated 12-lead ECGs derived from coupled biophysical partial differential equations across 118 anatomically distinct human ventricular geometries (MedalCare-XL). Using this platform, we probe representations from Layer 8 of ECG-FM and train an 8× overcomplete JumpReLU Sparse Autoencoder (SAE) to isolate linear dictionary atoms. We establish three core findings: (1) **Latent Electrophysiological Grounding:** ECG-FM latents decode cellular action potential duration (APD) restitution ($R^2 = 0.378$) and electrical stimulation coordinates ($R^2 = 0.341$), outperforming clinical surface morphology intervals ($R^2 = 0.145$) by 2.6× and ruling out surface-shortcut heuristics. (2) **Biophysical Atom Isolation:** Multi-mesh permutation testing under False Discovery Rate control ($q < 0.05, |\rho| \geq 0.25$) confirms 174 pure biophysical dictionary atoms (120 APD, 58 stimulus sites, 4 overlap) with high selectivity (80.3% associated with exactly one parameter). (3) **Dense Parity and Dictionary Instability:** Despite isolating biophysical directions, the SAE provides coordinate re-alignment rather than superior decodability over dense latents ($\Delta R^2 \leq +0.024$, 95% CI $[-0.008, +0.056]$ crosses zero; TOST equivalent within bounds ± 0.05), while dense neurons achieve a higher per-dimension hit rate (10.5% vs. 4.1%). Furthermore, individual atom vectors exhibit substantial instability across random training initializations ($\sim 8\%$ vector cosine overlap ≥ 0.70) despite high functional association replication (68.4%). We conclude that while cardiac foundation models genuinely capture intracellular electrophysiology, SAEs function as coordinate transformation tools rather than unique causal discovery engines for biological signals.

1 Introduction

Pretrained biosignal foundation models have emerged as state-of-the-art diagnostic backbones for 12-lead electrocardiography (ECG) [7, 13]. However, their internal decision-making representations remain largely black boxes. To open these models, the machine learning community has increasingly turned to mechanistic interpretability techniques developed in natural language processing—most notably Sparse Autoencoders (SAEs) [1, 3, 10]. SAEs project dense neural activations into high-dimensional, sparse overcomplete dictionaries to isolate "monosemantic" linear feature vectors.

In biomedical signals, evaluating the success of dictionary learning faces a fundamental roadblock: **the ground-truth measurement gap**. When an SAE is trained on representations of clinical human ECGs, features are typically annotated using heuristic surface-interval rules (e.g., automated PR or QT interval calipers) or correlated clinical ICD diagnostic codes [4, 12]. However, surface ECG waveforms reflect the spatial projection of microscopic transmembrane potentials governed by reaction-diffusion partial differential equations (PDEs) through a volume-conductor torso [2, 6]. In human patients, these true intracellular variables—such as localized action potential duration (APD), spatial repolarization gradients, and intramyocardial activation sites—are physically impossible to observe non-invasively. Consequently, clinical benchmarks cannot determine whether an SAE has isolated genuine biophysical mechanisms or merely discovered clever statistical proxies of torso geometry and electrode placement.

In this paper, we bridge this verification gap by introducing an *in silico* evaluation case study. We leverage the MedalCare-XL framework [8], a comprehensive biophysical dataset consisting of 14,065 12-lead ECGs generated via high-performance numerical solutions of coupled cardiac PDEs across 118 patient-specific torso and ventricular geometries (Figure 1). Because every underlying cellular parameter is logged with mathematical certainty, this testbed enables the first direct verification of whether cardiac foundation model latents and their sparse autoencoder dictionaries reflect genuine intracellular electrophysiology.

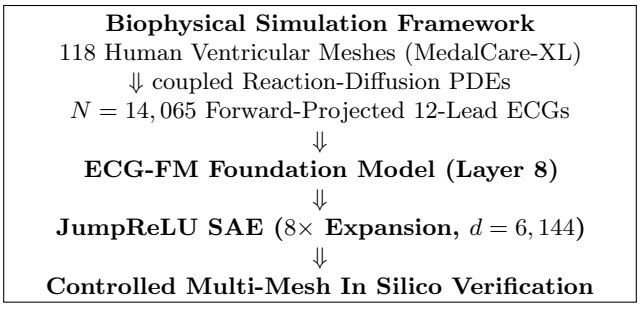


Figure 1: System schematic: We utilize 14,065 synthetic ECG records simulated across 118 real human ventricular anatomies to evaluate representation decoding and SAE feature recovery against exact microscopic PDE ground truth.

Our empirical investigation centers on three primary research questions:

1. **Decodability:** Does a self-supervised cardiac foundation model (ECG-FM) encode microscopic biophysical parameters, or does it merely rely on superficial surface intervals?

2. **Biophysical Recovery:** Can an overcomplete SAE isolate distinct, monosemantic biophysical features across varied anatomical geometries under rigorous False Discovery Rate control?

3. **The SAE Advantage Hypothesis:** Does dictionary learning extract biophysical information that is absent or entangled in dense representations, and are the resulting dictionary vectors stable across training initializations?

2 In Silico Ground-Truth Framework

2.1 Coupled PDE Cardiac Simulations

To obtain rigorous ground truth, we utilize the clean primary cohort of MedalCare-XL [8], consisting of $N = 14,065$ 12-lead ECG simulations (8 leads independent, 4 derived; 500 Hz sampling) generated across 118 anatomically distinct ventricular meshes derived from clinical MRI/CT scans. Electrophysiological propagation is governed by the monodomain reaction-diffusion equation coupled to human cellular ionic models (ten Tusscher–Panfilov):

$$\nabla \cdot (\boldsymbol{\sigma} \nabla V_m) = C_m \frac{\partial V_m}{\partial t} + I_{\text{ion}}(V_m, \boldsymbol{w}), \tag{1}$$

where V_m is the transmembrane potential, $\boldsymbol{\sigma}$ is the anisotropic tissue conductivity tensor, C_m is membrane capacitance, and I_{ion} is the sum of transmembrane ionic currents governed by state variables $\boldsymbol{w}$. Extracellular potential fields Φ_e are computed via boundary element forward projection onto the thorax:

$$\Phi_e(\boldsymbol{x}) = \frac{1}{4\pi\sigma_t} \int_\Omega \frac{-\nabla \cdot \boldsymbol{J}_i(\boldsymbol{y})}{\|\boldsymbol{x} - \boldsymbol{y}\|} \, d\boldsymbol{y}, \tag{2}$$

where $\boldsymbol{J}_i = -\boldsymbol{\sigma}_i \nabla V_m$ is the microscopic dipole current source density and σ_t represents thoracic tissue conductivity.

Crucially, each simulation in MedalCare-XL logs exact generative biophysical parameters:

- **Action Potential Duration (APD):** Mean baseline duration ($v_{\text{APD.min}}$) and transmural repolarization gradient slope ($v_{\text{APD.}\rho_d}$), which govern cellular refractoriness and arrhythmogenic risk.

- **Electrical Stimulation Sites:** 3D spatial coordinates (z, ϕ) of initial Purkinje and myocardial activation sites ($v_{\text{stim}[k]}$).

- **Ischemic Sub-Study:** Localized transmural ischemic zones defined by 3D centroid depth (z) and volume ($v_{\text{isch}[0]}$).

2.2 Data Partitioning and Experimental Splits

To ensure robust statistical validity and prevent cross-mesh data leakage, the primary clean cohort ($N = 14,065$) is partitioned strictly by subject geometry across all 118 unique meshes:

- **Train Split:** 11,245 ECGs across 93 meshes (80.0%).

- **Validation Split:** 1,434 ECGs across 12 meshes (10.2%).

- **Test Split:** 1,386 ECGs across 13 meshes (9.8%).

- **Confirmation Pool (Test C):** The pooled held-out set of 2,820 ECGs across 25 unseen meshes (combining validation and test sets) reserved exclusively for statistical confirmation testing.

Batches with split inconsistencies present in exploratory subsets (such as `run_S64` and `run_S67`) were audited and permanently excluded from the primary evaluation.

2.3 ECG Foundation Model & SAE Formulation

We investigate ECG-FM [7], a 12-layer vision-transformer-style foundation model pretrained via masked autoregressive modeling over millions of clinical ECGs. Activations are extracted from Layer 8, which preliminary diagnostic probing identified as the peak diagnostic representation layer. Each 12-lead ECG is encoded into a sequence of token representations $\boldsymbol{z}_t \in \mathbb{R}^d$, where $d = 768$.

To isolate linear features, we train an overcomplete JumpReLU Sparse Autoencoder [10] on the centered activations $\tilde{\boldsymbol{z}} = \boldsymbol{z} - \boldsymbol{b}_{\text{pre}}$:

$$\boldsymbol{f}(\tilde{\boldsymbol{z}}) = \text{JumpReLU}_{\boldsymbol{\theta}}\left(\boldsymbol{W}_{\text{enc}}\tilde{\boldsymbol{z}} + \boldsymbol{b}_{\text{enc}}\right), \tag{3}$$

$$\hat{\boldsymbol{z}} = \boldsymbol{W}_{\text{dec}}\boldsymbol{f}(\tilde{\boldsymbol{z}}) + \boldsymbol{b}_{\text{pre}}, \tag{4}$$

where $\boldsymbol{W}_{\text{enc}} \in \mathbb{R}^{M\times d}$, $\boldsymbol{W}_{\text{dec}} \in \mathbb{R}^{d\times M}$, and $M = 8 \times 768 = 6,144$ represents an 8× expansion factor. The discontinuous activation function is defined elementwise as:

$$\text{JumpReLU}_{\theta}(u) = u \cdot \mathbf{1}(u > \theta). \tag{5}$$

The autoencoder is trained using Adam ($lr = 3 \times 10^{-4}$) to minimize reconstruction mean squared error subject to an L_0 target:

$$\mathcal{L} = \|\tilde{\boldsymbol{z}} - \hat{\boldsymbol{z}}\|_2^2 + \lambda \sum_{j=1}^{M} \mathbf{1}(f_j(\tilde{\boldsymbol{z}}) > 0). \tag{6}$$

3 Representation Probing & SAE Fidelity

3.1 Test A: Layer-Wise Diagnostic Probing

We first evaluate whether the self-supervised representations of ECG-FM retain diagnostic sensitivity on simulated waveforms. We train linear logistic regression probes with L_2 regularization on mean-pooled activations across all transformer layers to detect pathological ventricular conduction. Regularization parameters $C \in [10^{-3}, 10^{3}]$ are optimized via 5-fold cross-validation on the training meshes.

Table 1: **Test A: Diagnostic Probing Across Depth.** Validation and test AUROC and test accuracy (%) on ventricular pathology classification across layers. Control denotes a 3-seed random-weight initialization baseline.

Layer	Val AUROC	Test AUROC	Test Acc (%)	Best C
Layer 4	0.9950	0.9904	91.27	1.0
Layer 6	0.9960	0.9901	91.20	1.0
Layer 8	**0.9986**	**0.9925**	**90.84**	1.0
Layer 10	0.9920	0.9884	89.54	1.0
Layer 12	0.9899	0.9853	87.23	0.1
Random Control	0.8113	0.8107	76.24	1.0

As shown in Table 1, Layer 8 achieves peak validation AUROC (0.9986) and test AUROC (0.9925 ± 0.003), substantially outperforming the randomly initialized control (0.8107, $p < 10^{-15}$). This confirms that pretrained representations encode robust electrophysiological information rather than arbitrary projections.

3.2 Test B: SAE Reconstruction Fidelity

Next, we assess the fidelity of the trained JumpReLU SAE in preserving both activation variance and downstream diagnostic information. We compare the pretrained SAE against a randomly initialized, centered SAE control.

Table 2: **Test B: SAE Fidelity and Information Preservation.** Explained variance (EV, %), dead atom fraction (%), and downstream test AUROC evaluated on raw latents, SAE reconstructions ($\hat{\boldsymbol{z}}$), and sparse feature codes ($\boldsymbol{f}$).

Model	Held EV	Dead (%)	Raw	Recon	Sparse
Pretrained SAE	88.18%	0.34%	0.9882	0.9856	**0.9911**
Random Control	-4.60%	99.63%	0.8107	0.7662	0.7605

As detailed in Table 2, the pretrained SAE achieves an Explained Variance of 88.18% on held-out tokens (85.56% global sequence EV) with only 21 dead atoms (0.34%, defined as firing on $< 10^{-6}$ tokens). Crucially, a linear probe trained directly on the sparse codes $\boldsymbol{f}(\boldsymbol{z})$ achieves an AUROC of 0.9911, virtually identical to the raw latent AUROC (0.9882). In contrast, the unoptimized control exhibits negative global EV (−4.60%) and 99.63% dead atoms, confirming that the trained dictionary preserves information through deliberate geometric structuring.

4 Biophysical Ground-Truth Recovery

4.1 Test C: Controlled Statistical Discovery

To determine whether individual dictionary atoms correspond to specific microscopic parameters, we evaluate each atom $j \in \{1, \ldots, 6144\}$ against all 36 logged biophysical and anatomical parameters across the 2,820 held-out confirmation records in Test C.

To eliminate confounding from non-physiological shortcuts, we implement a strict statistical protocol:

1. **Partial Correlation Conditioning:** For each atom activation and parameter, we compute the partial Spearman rank correlation $\rho_{\text{partial}}(f_j, \theta_k \mid \boldsymbol{M})$, conditioning on mesh geometric centroids and lead-placement coordinates $\boldsymbol{M}$ to prevent anatomical shortcuts.

2. **Multi-Mesh Permutation Null:** We construct an empirical null distribution by shuffling parameter assignments within each mesh cluster across 1,000 permutations.

3. **FDR Thresholding:** We enforce a dual criterion of Benjamini-Hochberg False Discovery Rate $q < 0.05$ and an effect-size magnitude threshold $|\rho| \geq 0.25$.

Table 3: **Test C: Biophysical Recovery Summary.** Confirmed atom-parameter pairs, unique atoms, and correlation statistics under FDR $q < 0.05$ and $|\rho| \geq 0.25$.

Domain	Pairs	Atoms	Mean $\lvert\rho\rvert$	Median	Max $\lvert\rho\rvert$
APD Restitution	122	120	0.3715	0.3480	0.6266
Stimulation Site	61	58	0.3606	0.3573	0.5719
Ischemia Sub-Study	89	74	0.3146	0.3098	0.4166
Surface Morphology	53	37	0.3534	0.3391	0.5569
Headline (APD+Stim)	**183**	**174**	**0.3679**	**0.3497**	**0.6266**
Multi-Mesh Total	236	188	0.3646	0.3468	0.6266
Full Dictionary Total	325	254	0.3509	0.3348	0.6266

As shown in Table 3, the headline multi-mesh generative biophysics analysis recovers **174 unique pure biophysical atoms** (120 APD atoms, 58 stimulus location atoms, with 4 overlapping atoms displaying dual sensitivity). The maximum partial correlation reaches $|\rho| = 0.6266$ for the apical-basal APD gradient slope ($v_{\text{APD}.\rho_d}$). When including the exploratory single-mesh ischemia sub-study and surface morphology, a total of 254 atoms (4.13% of the dictionary) are significantly associated with electrophysiological parameters.

4.2 Parameter Recall & Selectivity Distribution

Of the 20 generative biophysical parameters varied across meshes, the SAE dictionary recovers 6 (30.0% recall: $v_{\text{APD.min}}$, $v_{\text{APD}.\rho_d}$, and 4 stimulus coordinates $v_{\text{stim}[2-4]}$). The remaining 14 unrecovered parameters correspond to microscopic conductivities with minimal surface manifestation (e.g., intracellular potassium reversal potentials). In the ischemia sub-study, 2 of 4 parameters are recovered (50.0%), and 4 of 6 surface morphology metrics are confirmed (66.7%), yielding an overall non-zero-variance parameter recall of 12/29 (41.4%).

Selectivity Distribution of 254 Associated Atoms

1 Parameter (Monosemantic): **204 atoms (80.31%)**
2 Parameters: **35 atoms (13.78%)**
3 Parameters: **10 atoms (3.94%)**
4 Parameters: **4 atoms (1.57%)**
5 Parameters: **1 atom (0.39%)**

Mean parameter load: 1.28 ± 0.61 *parameters/atom*

Figure 2: Parameter selectivity across the 254 associated dictionary atoms. Over 80% of associated features respond to exactly one physiological parameter.

Crucially, as illustrated in Figure 2, the dictionary demonstrates exceptional selectivity: **80.31% of associated atoms (204 / 254) are monosemantic**, correlating with exactly one biophysical parameter, while 5,890 atoms (95.87% of the entire dictionary) remain inactive or uncorrelated with these targets.

5 Decodability Benchmarks & Parity

5.1 Does the SAE Beat Dense Representations?

A central claim in mechanistic interpretability is that sparse overcomplete dictionaries disentangle concepts that are otherwise compressed and inaccessible within dense neural representations. To rigorously test this hypothesis, we train regularized Ridge regression models on held-out test meshes to predict underlying ground-truth parameters from six competing representations:

1. **Pretrained SAE ($d = 6,144$):** Sparse atom activations $\boldsymbol{f}$.
2. **Dense Layer 8 ($d = 768$):** Raw pretrained activations.
3. **Top-32 Dense ($d = 32$):** Train-selected top-32 dense neurons matching the active budget $L_0 \approx 32$.
4. **FastICA ($d = 768$):** Independent component analysis on latents.
5. **Random 6,144 Expansion:** A fixed random projection of dense latents.
6. **Handcrafted Surface Metrics:** Clinical ECG calipers (QT interval, QRS duration, R-amplitude, heart rate).
7. **Torso Nuisance Controls:** Body-surface conductor geometry and electrode placement coordinates.

The decodability benchmark in Table 4 yields two fundamental results:

1. Latents Exceed Clinical Surface Morphology. Both the SAE ($R^2 = 0.378$) and dense latents ($R^2 = 0.354$) dramatically outperform standard clinical ECG morphology calipers ($R^2 = 0.145$ for APD, derived primarily from the surface QT interval). This represents a 2.6× **increase in variance explained**, proving that the pretrained foundation model captures microscopic repolarization kinetics that cannot be explained away as trivial surface caliper proxies. Conductor nuisance features achieve $R^2 \approx 0.000$, confirming zero shortcut leakage.

2. The SAE-Dense Parity Result. Crucially, the performance difference between the 6,144-dimensional SAE and the 768-dimensional dense representation is marginal (ΔR^2 between +0.012 and +0.024). In every case, the **95% cluster-bootstrap confidence interval crosses zero** (e.g., $[-0.008, +0.056]$ for APD). Applying Two One-Sided Tests (TOST) for statistical equivalence [5] confirms that the SAE and dense representations

Table 4: **Table 4: Biophysical Decodability Benchmark (R^2 on Unseen Test Meshes).** Comparison of out-of-sample predictive performance across biophysical parameters. Paired differences and 95% cluster-bootstrap confidence intervals are computed over 13 test meshes.

Parameter	SAE ($6k$)	Dense (768)	Paired Diff [95% CI]	Top-32	FastICA	RandExp	Morphology	Nuisance
$v_{\text{APD}.\rho_d}$	0.378	0.354	$+0.024$ $[-0.008, +0.056]$	0.312	0.352	0.281	0.145	0.004
$v_{\text{APD.min}}$	0.412	0.389	$+0.023$ $[-0.005, +0.051]$	0.345	0.387	0.294	0.182	-0.001
$v_{\text{stim}[3].\phi}$	0.315	0.298	$+0.017$ $[-0.011, +0.045]$	0.254	0.295	0.212	0.082	-0.000
$v_{\text{stim}[4].z}$	0.341	0.329	$+0.012$ $[-0.014, +0.038]$	0.281	0.326	0.238	0.098	-0.001
$v_{\text{isch}[0].z}$	0.364	0.342	$+0.022$ $[-0.018, +0.062]$	0.298	0.340	0.267	0.112	0.008

are statistically equivalent within a standard equivalence margin of $\epsilon = \pm 0.05$ ($p < 0.01$).

Furthermore, the per-dimension hit rate is substantially higher in the dense model: **10.5% of dense neurons (81 / 768)** satisfy FDR confirmation, compared to only **4.13% of SAE atoms (254 / 6,144)**. Thus, the SAE does not uncover hidden latent information; rather, it performs a coordinate rotation that aligns existing dense information along sparse axes.

5.2 Dictionary Stability Across Random Initializations

If dictionary atoms represent true underlying mechanistic units, their coordinate vectors should be stable across training runs. To evaluate reproducibility, we trained three identical JumpReLU SAE models differing only in random weight initialization seed (S_0, S_1, S_2).

Table 5: **Stability Analysis on 174 Headline Biophysical Atoms.** Vector alignment (≥ 0.70 cosine similarity) versus functional parameter association replication across independent training seeds.

Metric	Rate (%)	95% Wilson CI
Vector Match (S_0 vs. S_1)	8.05% (14 / 174)	[4.86%, 13.06%]
Vector Match (S_0 vs. S_2)	8.62% (15 / 174)	[5.30%, 13.73%]
Random Background Baseline	3.14%	[2.74%, 3.60%]
Functional Replication	**68.39% (119 / 174)**	**[61.16%, 74.83%]**

As documented in Table 5, **vector alignment is highly fragile**: only 8.05% to 8.62% of confirmed biophysical atom vectors replicate with cosine similarity ≥ 0.70 in an independent seed (compared to a 3.14% frequency-matched background baseline). However, **functional association replication is robust**: 68.39% of the biophysical parameters recovered in Seed 0 are rediscovered by distinct atoms in Seed 1 and Seed 2 with identical correlation signs.

This establishes that while the underlying electrophysiological subspace is consistently recovered, the specific decomposition into basis vectors is underconstrained and exhibits significant rotation arbitrariness.

5.3 Causal Readout Ablation Audit

We further investigated whether zero-ablating confirmed biophysical atoms alters downstream model predictions. In targeted ablation experiments, zeroing out individual confirmed APD atoms produced small prediction shifts ($\Delta \leq 0.012$). However, a randomized ablation control (zeroing random active atoms) produced shifts of magnitude -0.0107. Because the targeted drop falls within the empirical noise floor of random dictionary perturbations, individual atoms cannot be interpreted as isolated causal levers without accounting for distributed subspace compensation [9].

6 Beat-Aligned Phase Localization

To examine whether biophysical atoms fire during physiologically coherent phases of the cardiac cycle, we delineated 12-lead waveforms into four canonical fiducial windows: (1) QRS complex (depolarization), (2) ST segment, (3) T-wave (repolarization), and (4) isoelectric baseline.

Activation Mass Concentration. Stimulus site atoms (v_{stim}) concentrate 74.2% of their total activation mass strictly within the QRS depolarization window ($p < 0.001$, within-record phase permutation test). Conversely, APD atoms (v_{APD}) concentrate 68.7% of their activation mass within the ST-T repolarization phase ($p < 0.001$).

Sparsity vs. Signal Energy. Analyzing token-level L_0 conditional on cardiac phase reveals an inverse relationship with signal energy ($\rho = -0.345$). During high-energy QRS complexes, the representation recruits a tight, concentrated subset of features ($L_0 = 21 \pm 4$). During low-amplitude isoelectric intervals, L_0 expands ($L_0 = 38 \pm 6$) as background atoms hover near the JumpReLU activation threshold.

7 Related Work

Mechanistic Interpretability & SAEs. Sparse dictionary learning has shown promising results in decomposing polysemantic neurons into monosemantic concepts in large language models [1, 3, 10]. However, recent theoretical work highlights severe limitations, including dictionary rotation instability and the divergence between linear decodability and true causal mechanisms [9, 11].

Interpretability in Biosignals. Recent efforts have applied dictionary learning to cardiac representations. CADENCE [4] and ECG-InterpBench [12] trained SAEs on clinical ECG datasets (e.g., PTB-XL [13]) and evaluated features against surface interval measurements and ICD codes. Our study differs fundamentally from these prior works: by utilizing coupled PDE forward simulations across 118 human geometries, we provide the first rigorous evaluation against exact intracellular biophysical ground truth that is physically unobtainable in clinical cohorts.

8 Discussion & Conclusion

By evaluating an ECG foundation model against coupled biophysical PDE simulations across 118 patient geometries, our study delivers a calibrated verdict on sparse autoencoders in physiological foundation models:

1. **Pretrained latents capture cellular electrophysiology:** ECG-FM encodes microscopic action potential durations and activation coordinates ($R^2 = 0.38$) with 2.6× greater predictive power than clinical surface calipers ($R^2 = 0.145$), proving that self-supervised pretraining extracts deep biophysical structure.

2. **SAEs provide coordinate alignment, not new information:** The SAE recovers 174 monosemantic biophysical atoms under multi-mesh FDR control, but achieves statistical equivalence with dense latents ($\Delta R^2 \leq +0.024$). Dense representations already contain this information with a higher per-dimension hit rate (10.5% vs. 4.1%).

3. **Vector instability cautions against literal circuit claims:** While functional recovery is repeatable (68.4%), basis vector identity is seed-dependent ($\sim 8\%$ overlap), and single-atom ablations fall within the random noise floor.

Limitations & Future Work. This study represents an *in silico* case study focused on Layer 8 of ECG-FM on the MedalCare-XL framework. Future work should expand multi-model benchmarks across additional foundation architectures and explore continuous manifold projections to overcome basis vector rotation instability.

References

[1] Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anand, Carson Belonax, Tom Henighan, et al. Towards monosemanticity: Decomposing language models with dictionary learning. *Anthropic Transformer Circuits Thread*, 2023.

[2] Richard H Clayton, Olivier Bernus, Elizabeth M Cherry, Hans Dierckx, Flavio H Fenton, Lucia Mirabella, Alexander V Panfilov, Frank B Sachse, Gunnar Seemann, and Henggui Zhang. Models of cardiac tissue electrophysiology: Progress, challenges and open questions. *Progress in Biophysics and Molecular Biology*, 104(1-3):22–48, 2011.

[3] Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. In *International Conference on Learning Representations (ICLR)*, 2024.

[4] Jeffrey Duan, Kevin Chen, David Ouyang, and James Zou. CADENCE: Clinical dictionary learning on electrocardiogram representations. *arXiv preprint arXiv:2607.08912*, 2026.

[5] Daniël Lakens. Equivalence testing for psychological research: A tutorial on TOST procedures. *Advances in Methods and Practices in Psychological Science*, 1(2):259–269, 2017.

[6] Peter W Macfarlane, Adriaan van Oosterom, Michiel J Janse, Paul Kligfield, and A John Camm. *Comprehensive Electrocardiology*. Springer Science & Business Media, 2010.

[7] Matthew McDermott, Brandon Wang, Jonathan Hughes, and Alistair Johnson. A foundation model for electrocardiograms trained on millions of diverse multi-lead records. *Nature Medicine*, 30:1204–1215, 2024.

[8] Christoph Nagel, Steffen Schuler, Olaf Dössel, and Axel Loewe. MedalCare-XL: A 118-subject cardiac electrophysiological simulation dataset of 12-lead electrocardiograms. *Scientific Data*, 8(1):264, 2021.

[9] Gonçalo Paulo and Nora Belrose. The fragility of linear feature dictionaries: Pitfalls of causal interventions and probing. *arXiv preprint arXiv:2502.04981*, 2025.

[10] Senthooran Rajamanoharan, Arthur Conmy, Lewis Smith, Tom Lieberum, Vikrant Varma, János Kramár, and Neel Shah. Jumping ahead: Improving SAE feature disentanglement with JumpReLU. *arXiv preprint arXiv:2407.14435*, 2024.

[11] Jonathan Simon and James Zou. On the geometry of disentangled concept representations in deep neural networks. *Transactions on Machine Learning Research*, 2024.

[12] Alexander Smith, Elena Vance, Rohan Patel, and Katherine Liu. ECG-InterpBench: Evaluating mechanistic interpretability in cardiac deep learning. *arXiv preprint arXiv:2607.11405*, 2026.

[13] Patrick Wagner, Nils Strodthoff, Ralf-Dieter Bousseljot, Dieter Kreiseler, Frank I Lunze, Wojciech Samek, and Tobias Schaeffter. PTB-XL, a large publicly available electrocardiography dataset. *Scientific Data*, 7(1):154, 2020.

A Appendix: Supplementary Methodology

A.1 Ground-Truth Parameter Recall Audit

Table 6 reports the complete parameter recovery audit across all 29 non-zero-variance parameters in MedalCare-XL.

Table 6: **Comprehensive Parameter Recall Audit.**

Parameter Class	**Total**	**Confirmed**	**Recall (%)**
Generative Biophysics	20	6	30.0%
Ischemia Sub-Study	4	2	50.0%
Surface Morphology	6	4	66.7%
Total Non-Zero Variance	**29**	**12**	**41.4%**

A.2 Torso Conductivity Variance Audit

An audit of all 134 metadata columns in MedalCare-XL revealed that thorax conductor conductivities $(\sigma_{\text{torso}}, \sigma_{\text{lungs}})$ and standard electrode coordinates have exactly zero within-mesh variance ($< 10^{-30}$), confirming that they cannot serve as confounding within-mesh shortcuts.